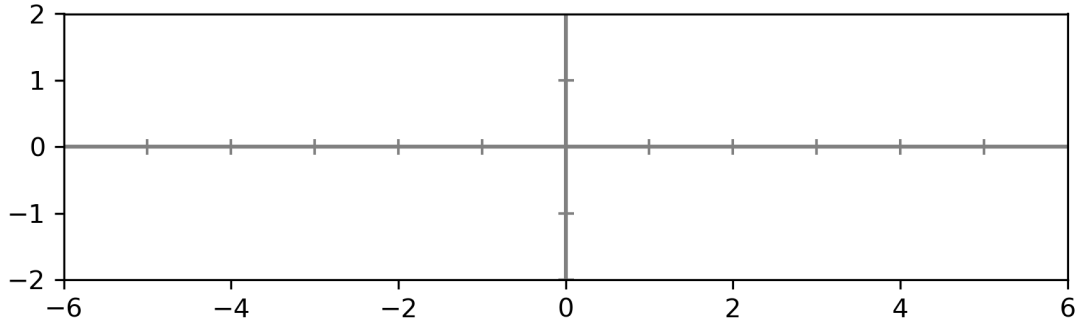
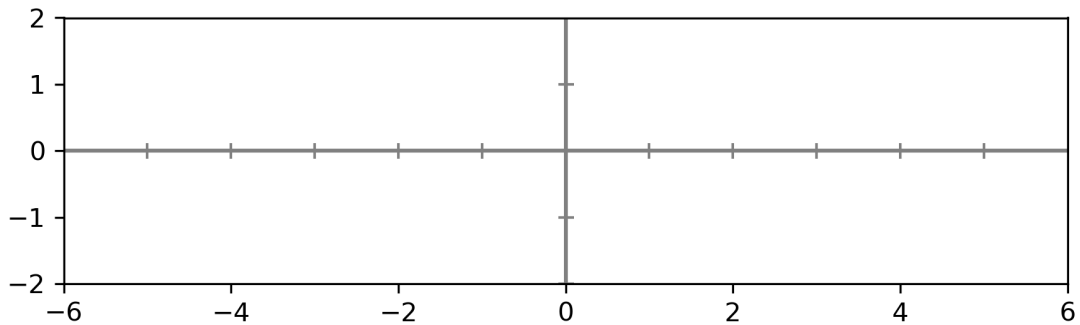


Signal Processing: Tutorial 1
AIMS-Rwanda, 2026

Q1) What does the Heaviside function look like? How would you shift it to the right by 2 units and scale it upward by 1.5 units?



Q2) Draw the standard cosine curve, of amplitude of 1, angular frequency of 1, and shifted to the left by $t = 2$. How would you right this analytically?



Q4) Plot each of these on the complex plane (Argand diagram, where the real axis is the “x” axis and the “y” axis is the imaginary axis). What is the magnitude and phase (in the range $[-\pi, \pi]$) of each?

a) $z_1 = 2e^{i\pi/4}$

b) $z_2 = -2e^{i\pi/4}$

c) $z_3 = 1.5$

d) $z_3 = 1.5i$

e) $z_4 = 4 - 2i$